

Mohammed Asim

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EDUCATION

University of California, Davis
Master of Science in Computer Science

Davis, CA
Sept. 2022 – March 2025

Jamia Millia Islamia
Bachelor of Technology in Computer Engineering

New Delhi, India
Aug. 2018 – June 2022

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript

Frameworks: Node.js, React, Express.js, Next.js, REST APIs

Databases: MongoDB, PostgreSQL, Redis, SQL

Developer Tools & Cloud: Docker, Kubernetes, GitHub, AWS (ECS2, Lambda, SageMaker)

Machine Learning & Data: TensorFlow, PyTorch, Keras, Pandas, NumPy, Scikit-learn, Deep Learning, Generative AI

EXPERIENCE

Graduate Student Researcher

June 2023 – Oct. 2024

UC Davis Health

Davis, CA

- Spearheaded the development of a high-performance Convolutional Neural Network (CNN) for MRI brain segmentation, achieving a Dice Similarity Score of 0.988, reducing manual annotation time by 70%.
- Implemented data preprocessing pipelines for a dataset of over 20,000 MRI scans, improving segmentation performance across varied head orientations by 25%.

Teaching Assistant (Machine Learning Course)

March. 2023 – June 2023

UC Davis

Davis, CA

- Led hands-on learning experiences for 70+ students in the Machine Learning course, boosting average engagement and comprehension scores by 10%.
- Graded and provided feedback on 50+ assignments to ensure students achieved an understanding of advanced Machine Learning and Data Analysis concepts.

Software Engineering Intern

June 2021 – July 2021

Airports Authority of India

New Delhi, India

- Developed a web-based application using the MERN stack to streamline inventory management, reducing reliance on paperwork by 50% and improving data accuracy across 4 departments.
- Delivered weekly progress reports and excel mockups to stakeholders, enhancing cross-departmental communication efficiency.

PROJECTS

Generative AI SQL Chat App | *Python, Streamlit, MySQL, SQLite, Groq API*

[Live App Link](#) | [GitHub](#)

- Built an AI-powered chatbot using the Groq API to translate natural language into SQL queries, enabling seamless interaction with SQLite and user-specified MySQL databases.
- Deployed an interactive Streamlit app with a secure UI for database configuration and real-time query execution.

Sleep Disorder Prediction using Machine Learning | *Flask, Python, Scikit-Learn, AWS EC2*

[Live App Link](#) | [GitHub](#)

- Designed a machine learning-driven web application to predict sleep disorders (Insomnia, Sleep Apnea, or None) using physiological and lifestyle inputs.
- Implemented a REST API and Flask web interface integrated with a Random Forest model, deploying the full-stack application on AWS EC2.

Personal Finance App | *MERN Stack, JWT, Vite, Render, Vercel*

[Live App Link](#) | [GitHub](#)

- Engineered a full-stack personal finance tracker with secure JWT authentication, RESTful APIs, and dynamic expense visualization using MongoDB, Express, React, and Node.js.
- Deployed the backend on Render and the frontend on Vercel, enabling real-time expense tracking with a responsive UI and protected user sessions.

PUBLICATIONS

- Scour Modeling Using Deep Neural Networks** | *ICT Express, Elsevier*
- Predicting Compressive Strength of Concrete** | *Cogent Engineering, Taylor & Francis*
- Predicting Next-Day Rainfall Using Machine Learning Techniques** | *(In Review)*

[Publication Link](#)
[Publication Link](#)
[Preprint](#)